A coarse sand barrier as an effective piping measure



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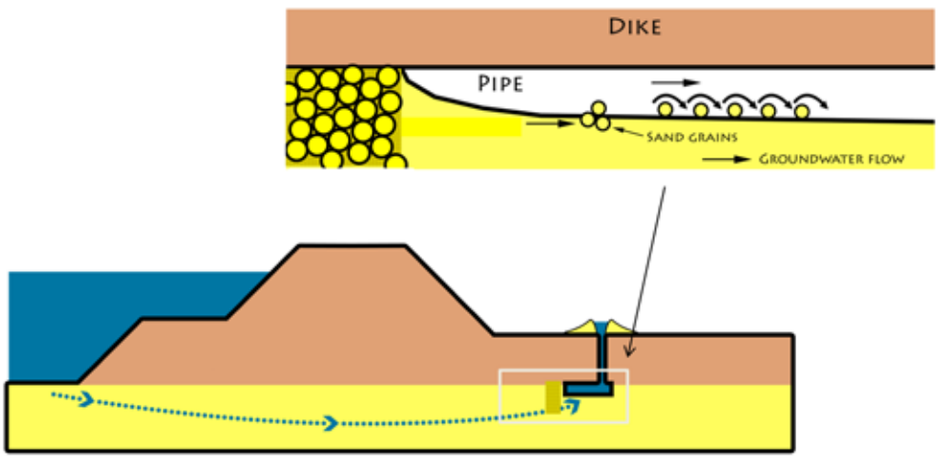


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Figure 1 - *The coarse sand barrier concept.*



Introduction

Backward erosion piping is a process whereby shallow pipes are formed at the interface of a cohesive layer and a sand layer, due to the removal of sand particles under the action of water flow. Ongoing pipe development can lead to severe erosion and finally failure of the water-retaining structure. The foundation that is susceptible to this mechanism, a combination of a uniform sandy layer covered by a cohesive layer is often encountered below river dikes in deltaic areas. Rise of the water level results in the formation of sand boils as a first sign of backward erosion. Numerous sand boils have been observed in the past, but failure due to backward erosion piping is not very common. Nevertheless, several dike failures in the Netherlands, China and the U.S. are attributed to this mechanism (Vrijling et al., 2010, Yao et al., 2009).

In the Netherlands, backward erosion piping is predicted with the Sellmeijer model (Sellmeijer, 1988, Sellmeijer et al. 2011, TAW, 1999). This model predicts the critical head on the basis of the groundwater flow towards the pipe, the viscous flow through the pipe and the limit-state equilibrium of particles at the pipe bottom. The model has been validated using experiments, but application in the field proves to be complex, as the required parameters are difficult to determine and show large fluctuation in the field. The uncertainty with respect to input parameters leads to the selection of conservative estimates, such that considerable reinforcements are due. The more stringent safety standards, due to a recent validation of the model, the inclusion of the length-effect and the risk approach, recently embraced in the Netherlands, lead to a further increase of the dike length to be reinforced.

Traditional measures, such as a berm, are not attractive when large seepage lengths are required and often houses are situated closely behind the dikes. Sheet pile walls are an alternative, but are economically unfeasible when it comes to application for long dike stretches.

Innovative or alternative piping measures are therefore becoming more and more popular. An example of an innovative measure is the vertical sand-retaining geotextile (Bezuijen et al., 2013, Förster et al., 2015). Using this method nearby the toe of the dike a vertical geotextile is inserted into a trench. Above the sand layer the trench is refilled with clay, such that upward seepage is not possible. An optimisation of this innovative solution is proposed here: a coarse sand barrier. In this solution the pipe formation is resisted by coarse sand instead of by the geotextile.

The concept

The coarse sand barrier relies on the concept that coarse sand provides more resistance to pipe formation than fine sand. The coarse sand is brought into the subsurface by creating a trench and simultaneous filling with coarse sand. Once a pipe forms, it will develop along the top of the sand bed and will collide with the barrier. As the coarse sand provides more resistance to erosion and the water flow is controlled by the overall properties of the aquifer, the water forces acting on the coarse particles are not sufficient for pipe development, unless the head across the structure is raised.

Abstract

Recent safety studies indicate a significant risk of backward erosion piping as a failure mechanism for dikes. Traditional measures, such as berms or sheet pile walls take a lot of space or are expensive, in particular in areas with a high infrastructural density. Consequently, there is an urgent need for alternative piping measures. An innovative solution is the vertical sand-retaining geotextile, which has been developed and tested in the field. An optimisation of this method is a coarse sand barrier: a trench filled with coarse sand, covered by a clay layer. The coarse sand provides more resistance to erosion than the inherent sand of the aquifer with its uncertain composition with respect to d70, so that initiated pipes cannot continue to develop. Laboratory and field scale tests suggest that this is an effective and economically feasible method.

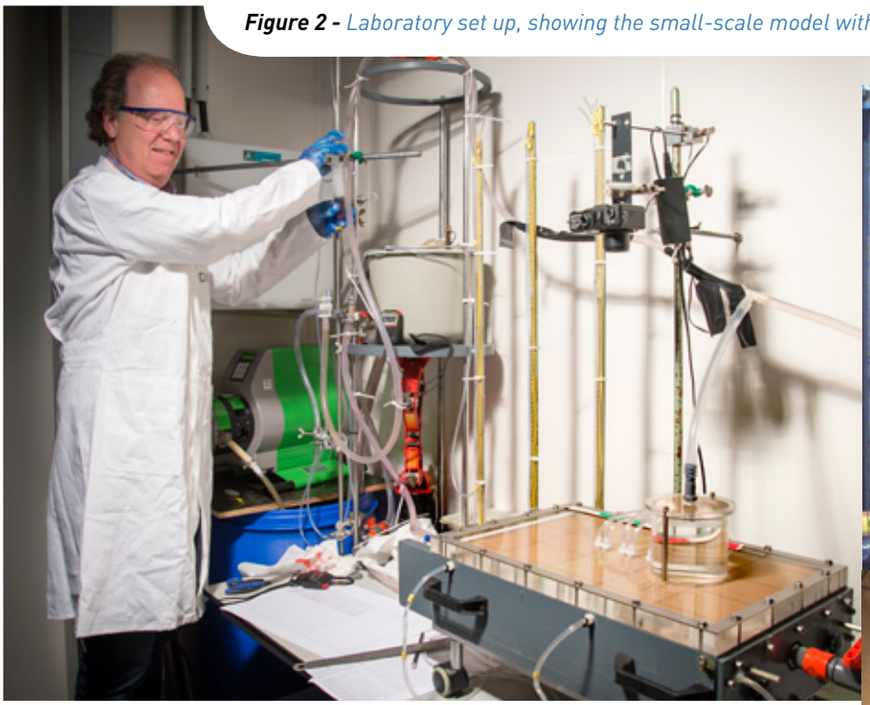


Figure 2 - *Laboratory set up, showing the small-scale model with circular exit.*

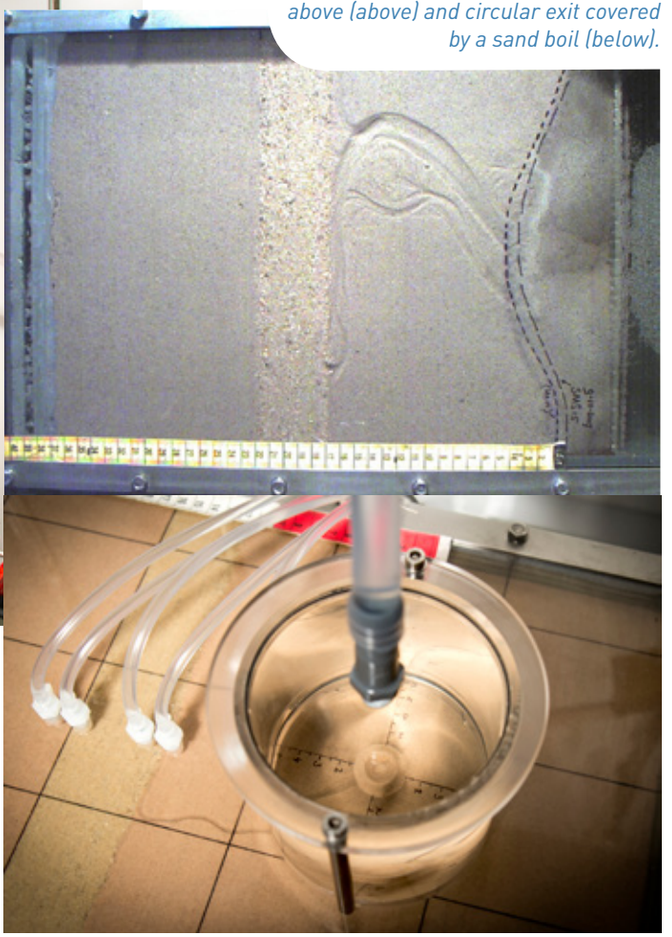


Figure 3 - *Slope type exit viewed from above (above) and circular exit covered by a sand boil (below).*

The permeable barrier will deflect excessive vertical seepage below the pipe tip, such that fluidisation of the sand bed below the pipe is less likely than for an impermeable structure like a sheet pile wall.

As in the method with the geotextile, above the sand the trench is filled with clay, to prevent upward seepage. Due to the clay filling above the barrier, the method is different from a more common filter, which aims for controlled discharge of water. For the coarse sand barrier the discharge is not expected to increase, not more than would be the case for any other barrier-type solution. Clogging is not expected, as long as the barrier is continuously below the water level.

Laboratory evidence

Laboratory experiments have been performed to illustrate the functioning and potential of a coarse sand barrier. A small-scale box (described in more detail in Van Beek et al., 2011 and Van Beek et al., 2015) with transparent cover (simulating the dike) was used for the experiments. Two configurations were used, one with an open exit representing a 2D exit with unconstrained flow towards the surface (also described in Van Beek et al., 2008) and one with a circular exit in the cover, representing a 3D exit with concentrated flow towards a single point (Figure 3).

In both configurations the box was filled with fine sand with a band of (medium) coarse sand and a head difference was applied to the sand until pipe formation occurred. The sand types used in the slope-type configuration were Playground sand and Masonry sand (with a d50 of 0.191 and 0.454 mm respectively) and the sand types used in the hole-type configuration were Baskarp sand and Itterbeck fraction 431 µm sand (with a d50 of 0.132 and 0.342 mm respectively). All sand types are uniform (d60/d10 1.5 – 2.6).

The observed process was similar in both configurations: the head drop was increased until a pipe formed. This pipe developed up to the coarse sand and started to develop parallel to the direction of flow and adjacent to the coarse sand barrier. Upon significant increase of head the pipe passes through the coarse sand barrier and develops towards the upstream side.

The critical hydraulic heads obtained in the experiments are indicated in Figure 4 for the samples with a coarse sand barrier and their homogeneous equivalents. It is noted that the upstream filter resistance causes quite some head loss in the slope-type configuration, such that the actual head loss across the sand bed is

smaller. Nevertheless, the difference between the critical heads in the experiments with and without sand barrier is very large, illustrating the potential of the method.

Field evidence

As the critical head of piping is affected by scale, it is relevant to also test the method at larger scale.

In 2009 and 2012 large-scale experiments were performed at the location of the IJkdijk. In these tests an actual dike was built on top of a sand bed placed in a basin. The test dike was 3.5 m high, 15 m long and 15 m wide at its base. It was constructed of compacted clay. The base consisted of a 3 m thick sand layer which extended 15 m beyond the test dike both at the upstream side and at the downstream side. Homogeneous tests without measures were performed in 2009. One of the purposes of the experiments in 2012 was to test piping measures, of which one of them was a coarse sand barrier. This coarse sand barrier was applied as an obstructing bar underneath the dike, 0.5 m wide and 0.5 m deep, at about one quarter of the seepage length from the downstream toe, see Figure 5 and 6. The coarse sand filter has been applied in the same basin where the first IJkdijk piping test had been carried out (Van Beek et al., 2011). In the upper 0.5 m a new, comparable sand was placed, with d50 =0.180 mm, Cu =1.7, d70 =0.207 mm. For the selection of a suitable sand for the filter, three criteria should be met: the filter should have a sufficient permeability, it should retain the finer material of the test sand and it should be internally stable. Based on filter criteria by Terzaghi and given by Giroud (2010) and Burenkova (1993) the coarse sand was selected, with d50=1.331 mm, Cu=1.3, d70=1.52 mm, d10=1.054 mm, d15=1.085 mm, d90=1.79 mm.

The first sand boil appeared at a head of 1.60 m. The head was increased until a level of 3.49 m, which equalled the height of the levee and was therefore the maximum head that could be applied in this experiment. In the homogeneous equivalent, performed in 2009, the dike failed at a hydraulic head difference of 2.1-2.3 m. These experiments indicate that at this scale the dike with the coarse sand barrier can at least withstand a head that is 1.6 times that of a dike without the barrier.

Discussion and Conclusions

The laboratory and field experiments illustrate that the application of a coarse sand barrier as a piping measure is promising. In the small-scale experiments an increase by a factor of 3-4 was established. An increase in strength of at least 1.6 is obtained in the field experiments at which failure due to piping did not even occur. Design rules should be based on filter criteria, heave criteria and the horizontal resistance against piping, which still requires investigation. Due to the experience with the vertical geotextile as a

Figure 4 - *Critical gradients obtained in the experiments.*

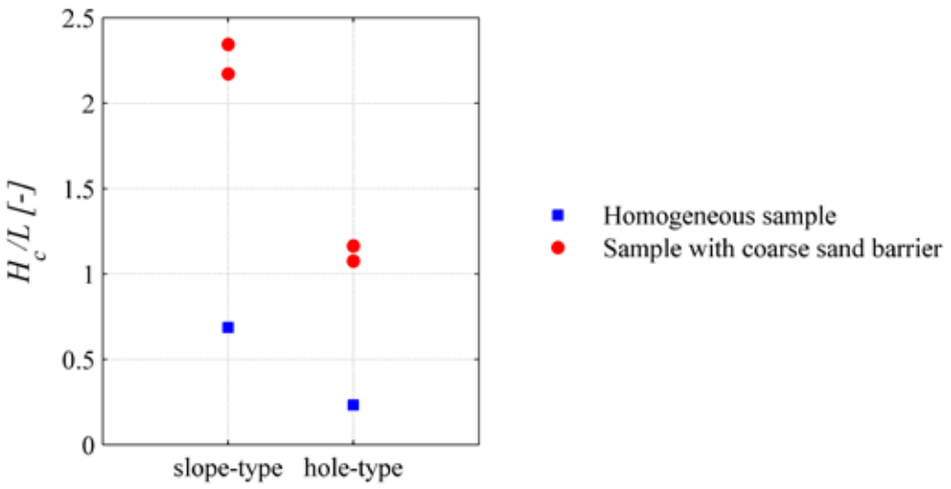


Figure 5 - *IJkdijk test schematisation showing the location of the coarse sand barrier.*

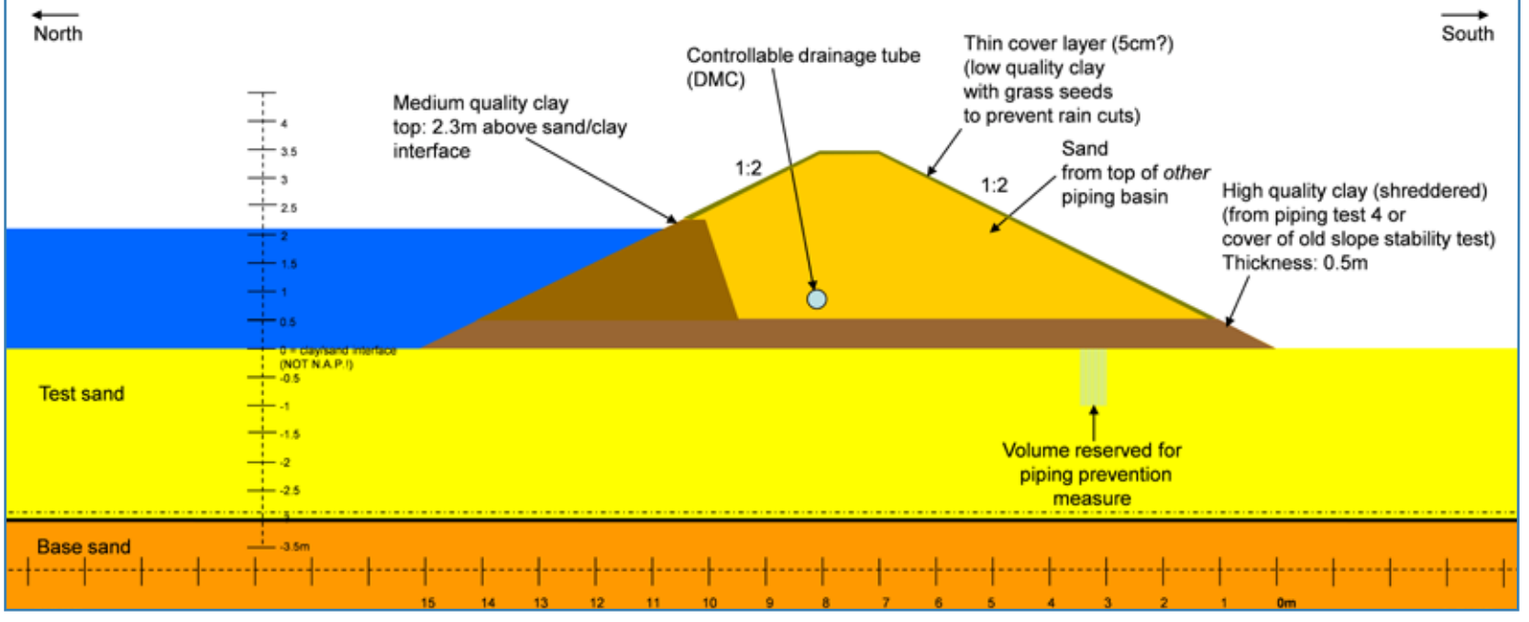




Figure 6 - *Digging the trench for the barrier (left) and the coarse sand barrier (right). At the background the dike that is to be placed is drawn at the side of the basin.*

piping measure, which is in many ways similar, practical issues for application in the field are likely to be resolvable.

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